

SIMATS
ENGINEERING

Secure Exit

MATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Debugging and Optimization

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QUESTIONS

Q1ATM Withdrawal – Exception
Handling Debugging
10 marks**Q2**Hospital Registration –
NullPointerException
Debugging
10 marks**Q3**E-Commerce Product Search –
Array Exception Debugging
10 marks**Q4**Railway Reservation –
Synchronization Debugging
10 marks**Q5**Bank Account – Race Condition
Debugging

Q1 ATM Withdrawal – Exception Handling Debugging

10 marks

An ATM application is used to process cash withdrawals. The following Java program contains programming errors.

Task:

1. Run the given program and identify the errors.
2. Debug and modify the program.
3. Submit the COMPLETE corrected Java program.
4. The corrected program must handle insufficient balance, zero withdrawal, invalid input, and unexpected errors.
5. Use appropriate try-catch-finally exception handling.
6. The corrected program must execute successfully for all valid test cases.

Faulty Program:

```
import java.util.Scanner;

class ATM {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
System.out.print("Enter account balance: ");
int balance = sc.nextInt();

System.out.print("Enter withdrawal amount: ");
int amount = sc.nextInt();

try {
    int remaining = balance / amount;

    if (amount > balance) {
        System.out.println("Insufficient balance");
    } else {
        System.out.println("Withdrawal successful");
        System.out.println("Remaining balance = "
            + (balance - amount));
    }
}

catch (ArithmeticException e) {
    System.out.println("Transaction failed");
}

System.out.println("Transaction completed");
}
```

Expected correction:

The program must correctly validate the withdrawal amount before performing the transaction, handle division-by-zero and invalid input, and always display "Transaction completed" after exception handling.

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String arg[]){
4         Scanner sc=new Scanner(System.in);
5         try{
6             int b=sc.nextInt();
7             int a=sc.nextInt();
8             if(a==0){
9                 throw new ArithmeticException("Withdrawal amount ");
10            }
11            if(a<0){
12                System.out.println("Invalid withdrawal amount");
13            }
14            else if(a>b){
15                System.out.println("Insufficient balance");
16            }
17            else{
18                int rb=b-a;
19                System.out.println("Withdrawal successful");
20                System.out.println("Remaining balance =" + rb);
21            }
22        }
23        catch(InputMismatchException e){
24            System.out.println("invalid input");
25        }
26        catch(ArithmeticException e){
27            System.out.println(e.getMessage());
28        }
29        catch(Exception e){
```

```
30         System.out.println("Transaction Failed");
31     }
32     finally{
33         System.out.println("Transaction completed");
34     }
35 }
36 }
```



Test Input

10000
3000



Output

Withdrawal successful
Remaining balance =7000
Transaction completed

🕒 Pending manual review

✅ Submitted

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